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 SHARDA UNIVERSITY <i>Beyond Boundaries</i>		School of Computing Science and Engineering Department of: Computer Science and Engineering				
Assignment-1, Session: 2026-2027 [Programme: B.Tech. CSE] [Semester: V]						
Course Title: Soft Computing Course Code: CSA301				Max Marks: 20		
Instructions:		1. All questions are compulsory 2. Assume missing data suitably, if any 3. All questions should be of higher learning.		BTL Level (K)		
Unit 1		CO1: Identify the need, significance, and fundamental principles of Soft Computing techniques for solving complex real-world problems.				
Unit 2		CO2: Define the basic concepts of Soft Computing.				
Q.No.	Section A (4×3=12 Marks)			Unit	COs	K
1	Define the four constituent paradigms of Soft Computing. For each, state in one line the type of imperfection it is designed to handle.			Unit 1	CO1	K1
2	Compare Hard Computing and Soft Computing on any eight parameters, presenting your answer as a table. Give one concrete application that is best solved by each.			Unit 1	CO1	K2
3	Define a fuzzy set and its membership function. State two differences between a crisp set and a fuzzy set, using the characteristic function in your answer.			Unit 2	CO2	K1
4	Define support, core, height, normality and α -cut of a fuzzy set. Illustrate all five on a single labelled trapezoidal membership function.			Unit 2	CO2	K2
Section B (2×4=8 Marks)						
5	Discuss any five societal applications of soft computing techniques (one each from healthcare, transport, consumer appliances, power systems and finance). For each, name the technique used and justify in two lines why hard computing is unsuitable.			Unit 1	CO1	K2
6	An aggregated output fuzzy set is given on the universe $y \in \{10, 20, 30, 40, 50\}$ with memberships $\mu = \{0.1, 0.5, 0.9, 0.4, 0.2\}$ respectively. Compute the defuzzified output by the center of gravity method.			Unit 2	CO2	K2